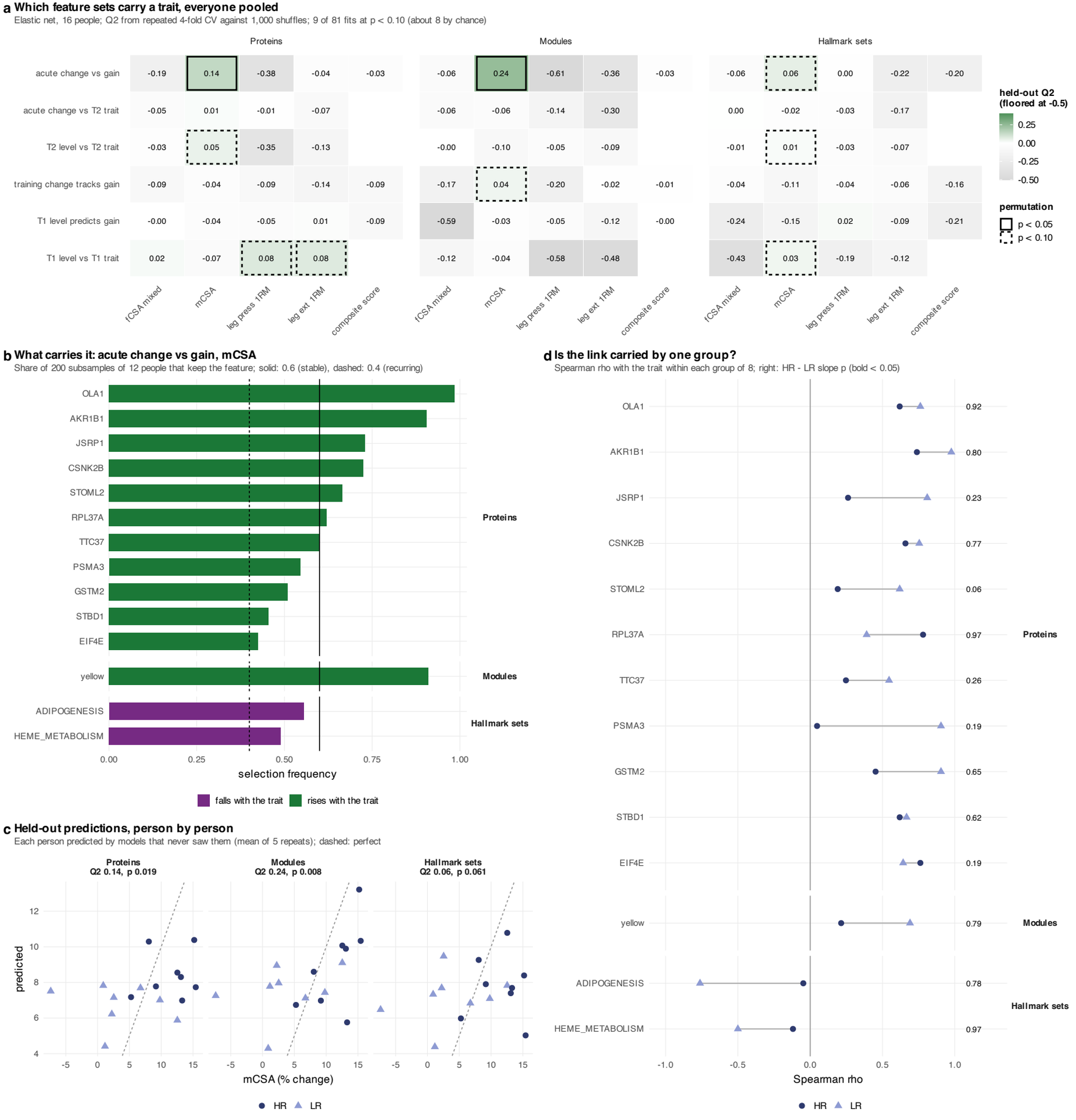
a Which feature sets carry a trait, everyone pooled
Elastic net, 16 people; Q2 from repeated 4-fold CV against 1,000 shuffles; 9 of 81 fits at p < 0.10 (about 8 by chance)
Proteins
Modules
Hallmark sets
acute change vs gain
acute change vs T2 trait
T2 level vs T2 trait
training change tracks gain
T1 level predicts gain
T1 level vs T1 trait
-0.19 0.14 -0.38 -0.04 -0.03
-0.05 0.01 -0.01 -0.07
-0.03 0.05 -0.35 -0.13
-0.09 -0.04 -0.09 -0.14 -0.09
-0.00 -0.04 -0.05 0.01 -0.09
0.02 -0.07 0.08 0.08
-0.06 0.24 -0.61 -0.36 -0.03
-0.06 -0.06 -0.14 -0.30
-0.00 -0.10 -0.05 -0.09
-0.17 0.04 -0.20 -0.02 -0.01
-0.59 -0.03 -0.05 -0.12 -0.00
-0.12 -0.04 -0.58 -0.48
-0.06 0.06 0.00 -0.22 -0.20
0.00 -0.02 -0.03 -0.17
-0.01 0.01 -0.03 -0.07
-0.04 -0.11 -0.04 -0.06 -0.16
-0.24 -0.15 0.02 -0.09 -0.21
-0.43 0.03 -0.19 -0.12
fCSA mixed
mCSA
leg press 1RM
leg ext 1RM
composite score
held-out Q2 (floored at -0.5)
0.25
0.00
-0.25
-0.50
permutation
p < 0.05
p < 0.10
b What carries it: acute change vs gain, mCSA
Share of 200 subsamples of 12 people that keep the feature; solid: 0.6 (stable), dashed: 0.4 (recurring)
OLA1
AKR1B1
JSRP1
CSNK2B
STOML2
RPL37A
TTC37
PSMA3
GSTM2
STBD1
EIF4E
yellow
ADIPOGENESIS
HEME_METABOLISM
Proteins
Modules
Hallmark sets
0.00
0.25
0.50
0.75
1.00
selection frequency
falls with the trait
rises with the trait
c Held-out predictions, person by person
Each person predicted by models that never saw them (mean of 5 repeats); dashed: perfect
Proteins
Q2 0.14, p 0.019
Modules
Q2 0.24, p 0.008
Hallmark sets
Q2 0.06, p 0.061
predicted
mCSA (% change)
HR
LR
d Is the link carried by one group?
Spearman rho with the trait within each group of 8; right: HR - LR slope p (bold < 0.05)
OLA1 0.92
AKR1B1 0.80
JSRP1 0.23
CSNK2B 0.77
STOML2 0.06
RPL37A 0.97
TTC37 0.26
PSMA3 0.19
GSTM2 0.65
STBD1 0.62
EIF4E 0.19
yellow 0.79
ADIPOGENESIS 0.78
HEME_METABOLISM 0.97
Proteins
Modules
Hallmark sets
Spearman rho
HR
LR